

SMART COOP PREDATOR DETERRENT

An ESP32-Based Automated Sensing and Deterrent System for Backyard Poultry Protection

[Team Member Name(s)] • [Institution / Department] • [Course / Event] • [Date]

PROJECT OVERVIEW

Small-scale poultry keepers regularly lose chickens to nighttime predators — the loss is sudden, largely unwitnessed, and hard to prevent with fencing alone. This project implements a self-contained, low-cost monitoring and deterrent unit built around an ESP32 microcontroller. The system automatically arms itself after dark using ambient light sensing, detects motion in the coop with a PIR sensor, and responds within a second with a bright light and audible siren — while also publishing live status and alerts over MQTT so an owner can monitor the coop remotely and trigger the deterrent manually from anywhere.

<1s

Motion-to-deterrent response time

2

Control modes — automatic (light-based) and manual override

6

Sensor & actuator subsystems integrated on one board

DESIGN GOALS

Fast, Automatic Response

Arm only after dusk (ambient-light sensing); detect intrusion and react without any human present.

Owner Override, Anywhere

A physical button for on-site testing, plus MQTT commands so the system can be armed, disarmed, or triggered remotely.

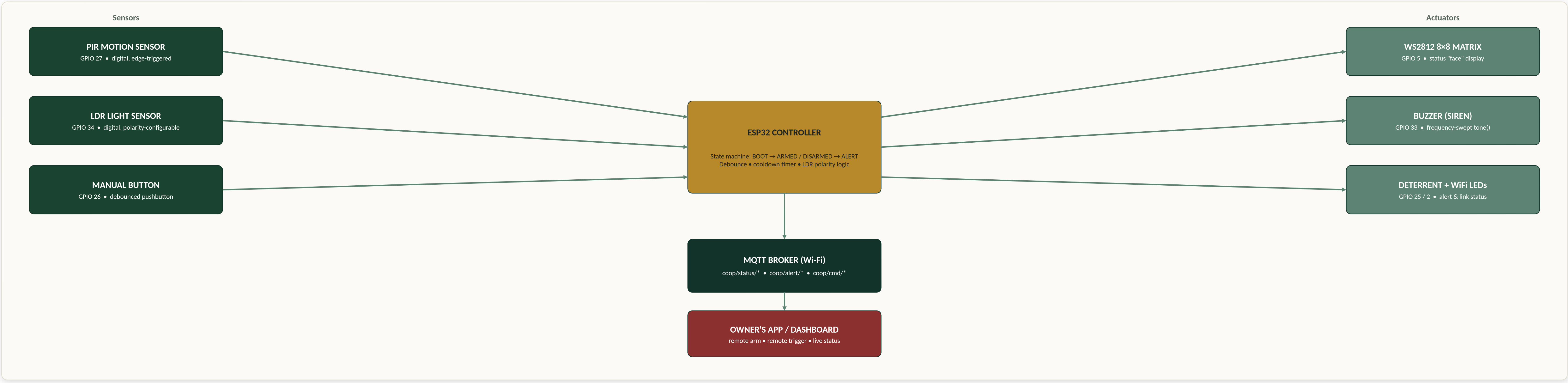
Bench-Testable in Isolation

Every subsystem — LED, siren, matrix, sensors — has its own serial test command, so hardware faults are diagnosed part-by-part, not by guessing.

Low-Cost, Repairable Hardware

Off-the-shelf ESP32, passive components, and a WS2812 matrix — no proprietary hardware, fully open firmware.

SYSTEM ARCHITECTURE



KEY FEATURES

1 Light-Based Auto-Arming

A comparator-output LDR module lets the coop arm itself at dusk and stand down at dawn — no schedule to set.

2 Edge-Triggered Motion Response

The PIR handler fires only on a LOW→HIGH transition while armed, with a 3-second cooldown to prevent alarm flooding.

3 Sweeping Siren + Strobing Alert

A frequency-swept tone (800–2400 Hz) pairs with a flashing matrix "eyes" glyph for maximum deterrent effect.

4 Remote Arm / Disarm / Trigger

MQTT commands (coop/cmd/arm, coop/cmd/deterrent) let an owner override behavior from anywhere with connectivity.

5 Live Status Telemetry

PIR, light, and armed state publish every 5 seconds; a retained "online" topic with Last-Will lets the app detect a dropped device.

6 Visual System "Face"

The 8x8 matrix shows a distinct glyph per state — spinner while booting, sun by day, breathing moon by night, flashing eyes on alert.

TECHNICAL SPECIFICATIONS

Hardware & Pin Map

Component	GPIO	Notes
PIR motion sensor	GPIO 27, digital in	3.3V logic only — verify module output
Manual button	GPIO 26, INPUT_PULLUP	Debounced, 50 ms window
LDR light module	GPIO 34, digital in	Comparator output, polarity configurable
WS2812 8x8 matrix	GPIO 5, DIN	64 pixels — VDD, GND, DIN required
Buzzer	GPIO 33	Passive tone() or active-module fallback
Deterrent LED (red)	GPIO 25	Digital out, dimmed on ack overlay
Wi-Fi status LED (green)	GPIO 2	Blink = connecting, solid = online

MQTT Topic Map

MQTT Topic	Direction	Payload / Purpose
coop/status/pir	publish	"0" / "1" — live motion state
coop/status/light	publish	"0" / "1" — raw LDR reading
coop/status/armed	publish	"0" / "1" — current arm state
coop/alert/predator	publish	"1" — fires once per trigger
coop/status/online	publish (LWT)	retained; "0" on disconnect
coop/cmd/deterrent	subscribe	"trigger" — force-fire remotely
coop/cmd/arm	subscribe	"auto" / "on" / "off"

TESTING, VALIDATION & FUTURE WORK

Bench-Test Methodology

- Serial test console isolates each subsystem before integration — led, buzzer2, matrix, status commands
- LDR polarity verified empirically per-unit (no fixed "dark = HIGH" assumption across comparator modules)
- Deterrent cooldown (3 s) and debounce (50 ms) tuned against false triggers on the physical bench unit
- Wiring and power faults (e.g. matrix requiring a common GND, not just VDD + DIN) documented and diagnosed live during build

Planned Extensions

- Reintroduce an automated door-lock servo, synced to the same dusk/dawn light logic
- Battery + solar power path for fully off-grid deployment
- Push notifications (rather than polling) on the owner's dashboard for predator alerts
- Data logging of trigger events for pattern analysis across seasons

Conclusion

The system demonstrates that a single low-cost microcontroller can combine light-aware arming, motion-triggered deterrence, and remote MQTT monitoring into one coherent, independently testable unit — a practical, extensible foundation for automated small-scale coop protection.